Chapter 5

Transformations of Random Variables and the Normal (Gaussian) Distribution

5.1 Transformations of random variables

Example 5.1. Suppose a 2kg robot is programmed to move along a straight track from point A to point B, the speed, X, of the robot when it reaches point B is recorded in meters per second. Due to variations in motor performance and friction we model the final speed as $X \sim \mathrm{U}(0.5, 1.5)$. When considering risks involved with crashes we work with the kinetic energy

$$Y = \frac{1}{2}mX^2$$

where m is the mass of the robot in kg. Calculate:

1. The expected final speed.
2. The expected final kinetic energy.
3. The probability that the final kinetic energy is less than 2.
4. Describe the distribution of the final kinetic energy.

Solution:

1. The expected final speed is $\mathbb{E}[X]$ and is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_{0.5}^{1.5} x dx = \left[\frac{1}{2}x^2\right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{2} = 1.$$

2. The expected final kinetic energy, $\mathbb{E}[Y]$, is given by

$$\mathbb{E}[Y] = \mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_{0.5}^{1.5} x^2 dx = \left[\frac{1}{3}x^3\right]_{0.5}^{1.5} = \frac{1.5^3 - 0.5^3}{2} = \frac{3.25}{3} \approx 1.0833.$$

   Warning: Note that $\mathbb{E}[g(X)]$ is <u>not</u> the same as $g(\mathbb{E}[X])$.

3. We can find probabilities of Y by first finding the cumulative distribution function, F_Y, of Y.

   $$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(X \leq \sqrt{y}) = F_X(\sqrt{y}).$$

   Recall

$$F_X(x) = \begin{cases} 0, & x < 0.5, \\ x - 0.5, & 0.5 \leq x < 1.5, \\ 1, & x \geq 1.5. \end{cases}$$

Therefore

$$F_Y(y) = \begin{cases} 0, & \sqrt{y} < 0.5, \\ \sqrt{y} - 0.5, & 0.5 \leq \sqrt{y} < 1.5, \\ 1, & \sqrt{y} \geq 1.5. \end{cases}$$

In particular, $F_Y(y) = \sqrt{y} - 0.5$ for $0.25 \leq y \leq 2.25$, hence $\mathbb{P}(Y \leq 2) = \sqrt{2} - 0.5 \approx 0.9142$.

4. From the cumulative distribution function of Y we see that Y is a continuous random variable, since F_Y is continuous and piecewise smooth, with probability density function

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2} y^{-\frac{1}{2}}, \quad 0.25 \leq y \leq 2.25$$

and $f_Y(y) = 0$ otherwise.

In many applications of probability theory we are interested in a function (or transformation) of a random variable X. Specifically, suppose X is a random variable and suppose $g : \mathbb{R} \to \mathbb{R}$ is a function. Then $Y = g(X)$ is *also* a random variable and we can ask:

- What is the expected value of $g(X)$?
- What is the probability $g(X) \leq y$ for some $y \in \mathbb{R}$? That is, what is the cumulative distribution function, F_Y, of Y?
- Can we determine the probability density function (if it exists)?

Typically it is easier to calculate the expected value $\mathbb{E}[g(X)]$ directly without first calculating the cumulative distribution function or probability density function so it is important to consider which of these tasks you need to do!

If we know the cumulative distribution function F_X (or probability density f_X) for X, how can we find the cumulative distribution function (or probability density) for the random variable $Y = g(X)$?

Example 5.2. Suppose X is a random variable with cumulative distribution function $F_X(x)$ and probability density function $f_X(x)$ given by:

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geqslant 0 \\ 0 & x < 0 \end{cases} \qquad f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geqslant 0 \\ 0 & x < 0 \end{cases}$$

where λ is a fixed, positive constant.
X is an example of an exponential random variable with parameter $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$. Notice that $\text{Im}X = [0, \infty)$.

Suppose $Y = g(X) = X^3$ (i.e. $g(x) = x^3$). Let's determine the cumulative distribution function F_Y for Y:

Step 1: The first step is to determine $\text{Im}Y$ (i.e. the set of values that $Y = X^3$ can take). Since we know that $\text{Im}X = [0, \infty)$ (i.e. $0 \leqslant X < \infty$), we have $0 \leqslant X^3 < \infty$, that is, $\text{Im}Y = [0, \infty)$ too.

Step 2: The next step is to determine the cumulative distribution function F_Y for all $y \in \mathbb{R}$.

Since $\text{Im}Y = [0, \infty)$, we know that for any $y < 0$, $F_Y(y) = \mathbb{P}(Y \leq y) = 0$.

Next, suppose that $y \geq 0$. To compute $F_Y(y)$ in this case:

$$\begin{aligned}
\text{For } y \geqslant 0,\ F_Y(y) &= \mathbb{P}(Y \leqslant y) && \text{(definition of } F_Y\text{)} \\
&= \mathbb{P}(X^3 \leqslant y) && \text{(since } Y = X^3\text{)} \\
&= \mathbb{P}(X \leqslant y^{1/3}) \\
&= F_X(y^{1/3}) && \text{(express in terms of } F_X\text{)} \\
&= 1 - e^{-\lambda y^{1/3}} && \text{(by substituting } y^{1/3} \text{ into } F_X\text{)}
\end{aligned}$$

So we have:

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 1 - e^{-\lambda y^{1/3}} & y \geqslant 0 \end{cases}$$

You can check that F_Y is continuous and differentiable (almost everywhere). Therefore Y also has a density f_Y which we can find by *differentiating* F_Y. So the probability density function for Y is given by:

$$f_Y(y) = F_Y^{'}(y) = \begin{cases} 0 & y < 0 \\ \frac{\lambda \exp\left(-\lambda y^{1/3}\right)}{3y^{2/3}} & y \geqslant 0 \end{cases}$$

Short cut: If you do not need to know an explicit formula for the cumulative distribution function F_Y of Y, you can find the probability density f_Y for Y without working out the cumulative distribution function F_Y. Since (from the algebra above) $F_Y(y) = F_X(y^{1/3})$ for $y \geqslant 0$, then for $y \geqslant 0$ we have

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}\left(F_X(y^{1/3})\right) \overset{chain\ rule}{=} F_X'(y^{1/3})\frac{\mathrm{d}}{\mathrm{dy}}(y^{1/3}) = f_X(y^{1/3}) \cdot \frac{1}{3y^{2/3}} = \frac{\lambda e^{-\lambda y^{1/3}}}{3y^{2/3}}$$

and $f_Y(y) = 0$ for $y < 0$ since $\mathrm{Im}Y = [0, \infty)$.

Warning: note that $Y = X^3$ but $f_Y(y) \neq \big(f_X(y)\big)^3$

Example 5.3. Suppose W is a random variable with density function given by

$$f_W(w) = \begin{cases} \frac{3w^2}{2} & -1 \leqslant w \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

So, $\mathrm{Im}W = [-1, 1]$ since $f_W(w) > 0$ when $-1 \leq w \leq 1$ (and the density is equal to 0 elsewhere).

Let $Z = W^2$. Find the probability density $f_Z(z)$ for Z.

Step 1. As in the previous example, the first step is to determine $\mathrm{Im}Z$. Since $-1 \leq W \leq 1$ (because $\mathrm{Im}W = [-1, 1]$), we must have $0 \leq Z \leq 1$ (since $Z = W^2$). It follows that $\mathrm{Im}Z$ = [0,1].

Step 2. Since the probability density f_Z is the derivative of the distribution function F_Z, the next step is to determine the cdf F_Z.

Since $\mathrm{Im}Z$ = [0, 1], we must have

$$\begin{aligned}
F_Z(z) &= 0 && \text{for } z < 0 \\
F_Z(z) &= \mathbb{P}(Z \leqslant z) = 1 && \text{for } z \geqslant 1
\end{aligned}$$

For $0 \leqslant z < 1$, we find an expression for F_Z in terms of F_W:

$$\begin{aligned} F_Z(z) \overset{\text{definition of df}}{=} \mathbb{P}(Z \leqslant z) &= \mathbb{P}(W^2 \leqslant z) && \text{(since } Z = W^2\text{)} \\ &= \mathbb{P}(-\sqrt{z} \leqslant W \leqslant \sqrt{z}) && \text{(solve the inequality } W^2 \leqslant Z \text{ for } W\text{)} \\ &= \mathbb{P}(-\sqrt{z} < W \leqslant \sqrt{z}) && \text{(since } W \text{ has a density and is a cts random variable)} \\ &= F_W(\sqrt{z}) - F_W(-\sqrt{z}) \end{aligned}$$

At this stage we have enough information to work out f_Z even though we have not worked out an explicit formula for either F_W or F_Z, as follows.

Step 3. The last step is to differentiate F_Z to obtain f_Z.

From Step 2 above, it is clear that if either $z < 0$ or $z > 1$, $F'(z) = f_z(z) = 0$.

For $0 < z < 1$, we differentiate both sides of the equation

$$F_Z(z) = F_W(\sqrt{z}) - F_W(-\sqrt{z})$$

to obtain

$$\begin{aligned} f_Z(z) = F_Z'(z) &= \frac{\mathrm{d}}{\mathrm{dz}}\big(F_W(\sqrt{z}) - F_W(-\sqrt{z})\big) \\ &= F_W'(\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} + F_W'(-\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} && \text{(chain rule)} \\ &= f_W(\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} + f_W(-\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} && \text{(since } F_W' = f_W\text{)} \\ &= \frac{3}{2}\sqrt{z} && \text{(using the given function } f_W\text{)} \end{aligned}$$

So we have:

$$f_Z(z) = \begin{cases} \frac{3}{2}\sqrt{z} & 0 \leqslant z \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

Linear transformations

Proposition 5.1. *Suppose that X is a random variable with pdf $f_X(x)$, and that $Y = aX + b$ for some $a, b \in \mathbb{R}$ with $a \neq 0$. Then Y has probability density function $f_Y(y)$ given by*

$$f_Y(y) = \begin{cases} \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) & \frac{y-b}{a} \in \mathrm{Im}X \\ 0 & \textit{otherwise} \end{cases}$$

Proof Since $Y = aX + b$, then $X = \frac{Y-b}{a}$, so that $\mathrm{Im}Y = \{y : \frac{y-b}{a} \in \mathrm{Im}X\}$.
Consider the case $a > 0$. For y such that $\frac{y-b}{a} \in \mathrm{Im}X$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leqslant y) \\ &= \mathbb{P}\left(aX + b \leqslant y\right) \\ &= \mathbb{P}\left(X \leqslant \frac{y-b}{a}\right) && \text{(since } a > 0\text{)} \\ &= F_X\left(\frac{y-b}{a}\right) \end{aligned}$$

$$\Rightarrow \qquad f_Y(y) = \frac{dF_Y}{dy} = \frac{d}{dy}F_X\left(\frac{y-b}{a}\right) = f_X\left(\frac{y-b}{a}\right) \times \frac{d}{dy}\left(\frac{y-b}{a}\right) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

as required. □

Exercise: Prove the above result for the case $a < 0$.

5.2 The Normal (Gaussian) distribution

Normal random variables arise in many ways in probability and statistics. We start with the definition of the *standard* normal random variable.

Definition
A random variable Z has the standard normal distribution if and only if its density φ is given by

$$\boxed{\varphi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2} \qquad \text{for } -\infty < z < \infty}$$

$\big(\text{so } \mathrm{Im}Z = (-\infty, \infty)\ \big)$

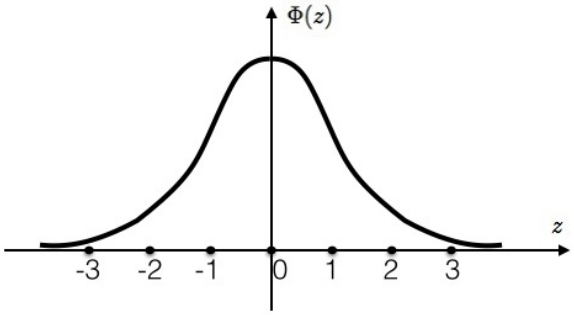


Figure 5.1: The graph of $\varphi(z)$

We will show that $\mathbb{E}(Z) = 0$ and $\mathbb{V}\mathrm{ar}(Z) = 1$. If Z has the standard normal distribution, we write $Z \sim \mathrm{N}(0, 1)$.

Remark 5.1. 1. The density φ is symmetric about $z = 0$, so $\mathbb{P}(Z \leqslant 0) = \mathbb{P}(Z \geqslant 0) = 1/2$

2. Since the density φ is symmetric about 0 it follows that $\mathbb{E}[Z] = 0$, indeed

$$\mathbb{E}[Z] = \int_{-\infty}^{\infty} z f_Z(z)\,\mathrm{d}z = \int_{-\infty}^{\infty} z\varphi(z)\,\mathrm{d}z = \int_{-\infty}^{0} \frac{ze^{-z^2/2}}{\sqrt{2\pi}}\,\mathrm{d}z + \int_{0}^{\infty} \frac{ze^{-z^2/2}}{\sqrt{2\pi}}\,\mathrm{d}z.$$

Now making the substitution $z \mapsto -z$ in the first integral

$$\mathbb{E}[Z] = \int_{0}^{\infty} \frac{-ze^{-z^2/2}}{\sqrt{2\pi}} dz + \int_{0}^{\infty} \frac{ze^{-z^2/2}}{\sqrt{2\pi}} dz = 0.$$

3. In general, it is not possible to express

$$\mathbb{P}(a \leqslant Z \leqslant b) = \int_a^b \frac{e^{-z^2/2}}{\sqrt{2\pi}}\mathrm{d}z$$

in terms of 'standard functions'.

Instead, we define a function $\Phi(x)$ to be

$$\Phi(x) = \int_{-\infty}^{x} \frac{e^{-z^2/2}}{\sqrt{2\pi}}\mathrm{d}z = \mathbb{P}(Z \leqslant x) \text{ for } Z \sim \mathrm{N}(0, 1)$$

4. **Using statistical tables**:
   Most calculators don't have a $\Phi(x)$ button, so instead we have to use statistical tables (or a computer package). Statistical tables give the values of the cumulative distribution function $\Phi(x)$, and then since $\mathbb{P}(a \leqslant Z \leqslant b) = \Phi(b) - \Phi(a)$, we can use the tables to compute probabilities for $Z \sim \mathrm{N}(0,1)$.

   Examples
   The Cambridge Statistical Tables give the values of $\Phi(x)$ for $x \geq 0$. So using the Cambridge Statistical Tables we can compute:

   $\mathbb{P}(Z \leqslant 1.2) = \Phi(1.2) = 0.8849$
   $\mathbb{P}(0.5 \leqslant Z \leqslant 1.2) = \Phi(1.2) - \Phi(0.5) = 0.8849 - 0.6915 = 0.1934$
   $\mathbb{P}(Z \geqslant 0.75) = 1 - \mathbb{P}(Z \leqslant 0.75) = 1 - \Phi(0.75) = 0.2266$

   Important: Using the symmetry of the distribution,

$$\Phi(-z) = \mathbb{P}(Z \leqslant -z) = \mathbb{P}(Z \geqslant z) = 1 - \mathbb{P}(Z \leqslant z) = 1 - \Phi(z)$$

   So:

$$\begin{aligned}\mathbb{P}(Z \leqslant -0.5) &= \Phi(-0.5) = 1 - \Phi(0.5) = 1 - 0.6915 = 0.3085\\ \mathbb{P}(-0.5 \leqslant Z \leqslant 0.75) &= \Phi(0.75) - \Phi(-0.5) = \Phi(0.75) - \big(1 - \Phi(0.5)\big)\\ &= \Phi(0.75) + \Phi(0.5) - 1 = 0.7734 + 0.6915 - 1 = 0.4649\end{aligned}$$

Proposition 5.2. *Let $Z \sim \mathrm{N}(0,1)$ be a standard Gaussian random variable then*

$$\mathbb{E}[Z] = 0$$
$$\mathbb{V}\mathrm{ar}[Z] = 1.$$

Proof As shown in Remark 5.1 the symmetry of the density gives that $\mathbb{E}[Z] = 0$ so it remains to show that $\mathbb{V}\mathrm{ar}[Z] = 1$.
Note that since $\mathbb{E}[Z] = 0$ we have $\mathbb{V}\mathrm{ar}[Z] = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = \mathbb{E}[Z^2]$ so it is sufficient to show that $\mathbb{E}[Z^2] = 1$.
We will show that $\mathbb{E}[Z^2] = 1$ by use of integration by parts. Using the expression for φ we can write

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \varphi(z)\,\mathrm{d}z = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z$$

Recall that the formula for integration by parts is

$$\int_a^b u(z)\,v'(z)\,dz = [u(z)v(z)]_a^b - \int_a^b v(z)\,u'(z)\,\mathrm{d}z.$$

We will apply this with $a = -\infty, b = \infty, u = z$ and $v = -e^{-\frac{z^2}{2}}$, noting that

$$v'(z) = \frac{d}{dz}\left(-e^{-\frac{z^2}{2}}\right) = ze^{-\frac{z^2}{2}}.$$

Then integration by parts gives

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z = \frac{1}{\sqrt{2\pi}}\left[-ze^{-\frac{z^2}{2}}\right]_{-\infty}^{\infty} - \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} -e^{-\frac{z^2}{2}}\,\mathrm{d}z.$$

Observe that the $ze^{-z^2/2} \to 0$ as $z \to -\infty$ or $z \to \infty$ therefore the boundary terms vanish and we have

$$\mathbb{E}[Z^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} \, \mathrm{d}z = \int_{-\infty}^{\infty} \varphi(z) \, \mathrm{d}z = 1.$$

The final equality holds since φ is a probability density function and hence integrates to 1. □

5.2.1 Linear transformation of $Z \sim \mathrm{N}(0,1)$

Let $Z \sim \mathrm{N}(0,1)$ and $X = \mu + \sigma Z$ for some $\mu \in \mathbb{R}$ and $\sigma \neq 0$. Then X has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Indeed, using Proposition 5.1 about linear transformations we have that if φ is the probability density function of Z then

$$f_X(x) = \frac{1}{|\sigma|}\varphi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{|\sigma|}\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

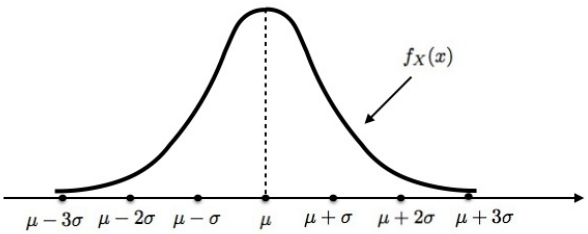


Figure 5.2: Sketch of $f_X(x)$

Observe the graph of f_X is bell-shaped and symmetric about $x = \mu$. We say that the function

$$\boxed{f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad -\infty < x < \infty}$$

is a normal density with *mean* μ and *variance* σ^2, and we write $X \sim \mathrm{N}(\mu, \sigma^2)$ to denote that X has a normal distribution with *mean* μ and *variance* σ^2 and density given by the function f_X above.

Proposition 5.3. *Let $X \sim \mathrm{N}(\mu, \sigma^2)$ be a Gaussian random variable then*

$$\mathbb{E}[Z] = \mu$$
$$\mathbb{V}\mathrm{ar}[Z] = \sigma^2.$$

Proof Let $X \sim \mathrm{N}(0,1)$ and write $X = \mu + \sigma Z$ for some $Z\mathrm{N}(0,1)$, by linearity of the expectation

$$\mathbb{E}[X] = \mathbb{E}[\mu + \sigma Z] = \mu + \sigma\mathbb{E}[Z] = \mu$$

since $\mathbb{E}[Z] = 0$. Moreover

$$\begin{aligned}\mathbb{V}\mathrm{ar}(X) &= \mathbb{V}\mathrm{ar}(\mu + \sigma Z) \\ &= \mathbb{V}\mathrm{ar}(\sigma Z) \\ &= \sigma^2 \mathbb{V}\mathrm{ar}(Z) \\ &= \sigma^2.\end{aligned}$$

Here we used the properties of the variance and that $\mathbb{V}\mathrm{ar}(Z) = 1$. □

Key fact: If $X \sim \mathrm{N}\left(\mu, \sigma^2\right)$, then $\frac{X-\mu}{\sigma} \sim \mathrm{N}\left(0,1\right)$. (This is called standardisation.)

Remark: We can use the Key Fact and the Cambridge Statistical Tables to calculate probabilities for X.

Often measurements of *physical quantities* follow a normal distribution.

Example 5.4. Based on medical data, researchers have concluded that the heights of 25-year old men follow a normal distribution with $\mu = 70$ inches and $\sigma^2 = 6.25$ in^2. If a 25-year old man is selected at random, what is the probability that he will be at least 6 feet tall?

Let X = height of randomly selected man. We need to compute

$$\begin{aligned}
\mathbb{P}(X \geqslant 72) &= \mathbb{P}\left(\frac{X-70}{\sqrt{6.25}} \geqslant \frac{72-70}{\sqrt{6.25}}\right) \\
&= \mathbb{P}(Z \geqslant 0.8) \qquad \left(\text{since if } X \sim \mathrm{N}\left(70, 6.25\right), \text{ then } Z = \frac{X-70}{\sqrt{6.25}} \sim \mathrm{N}\left(0,1\right)\right) \\
&= 1 - \Phi(0.8) = 1 - 0.7881 = 0.2119.
\end{aligned}$$